REFORM ALLIANCE · FEDERAL SENTENCING DATA ANALYSIS

Who Receives Federal Supervised Release?

What is linked to supervised release after a federal prison sentence, fiscal years 2014 to 2025

November 2026 · Supervised Release Analysis Report · Companion to the Probation Analysis Report

Supervised release is a period of supervision by a probation officer that follows a federal prison term. This report uses U.S. Sentencing Commission data to ask which people, offenses and places are linked to a higher or lower chance of receiving it. It compares people who are alike on the other characteristics in the data, and it checks whether the findings hold when missing information is handled in different ways.

0.049
odds ratio for non-U.S. citizens vs. U.S. citizens, the largest difference between people

31%
of the leftover variation is tied to state and district, almost all of it the district

1.57 → 1.14
odds ratio vs. FY2014, FY2024 peak and FY2025 drop

Key findings

1. **Citizenship shows the largest difference.** Non-U.S. citizens had 95% lower odds of supervised release than comparable U.S. citizens. The analysis links this to deportation after a prison term.
2. **Offense type comes next.** Sexual offenses had 4.4 times the odds of drug trafficking. Immigration and other offenses had about 84% to 85% lower odds, and fraud and violent offenses also had lower odds.
3. **Place matters a great deal.** About 31% of the variation left after accounting for people's characteristics is tied to the state and district. Nearly all of it is the district (28%), not the state (3.5%).
4. **The odds rose for a decade, then fell.** Compared with FY2014, the odds were higher in every later year, peaked in FY2024 (1.57) and fell to 1.14 in FY2025.
5. **Some results are not reliable.** Citizenship, ethnicity, gender and the FY2025 drop held up when missing data were handled differently. The education and firearms results changed or reversed.

Read this first. These results describe patterns in recorded sentences. They do not show why a difference exists or whether it reflects unequal treatment. The models leave out criminal history, the guideline range, plea agreements and the facts of each case, because the dashboard data do not include them.

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1 Purpose and data

The analysis was designed to identify what is linked to higher use of federal supervised release, to inform efforts to reduce unnecessary use. It looks for differences between demographic groups, between states and between judicial districts.

- **Data.** Federal sentencing records from the U.S. Sentencing Commission, fiscal years 2014 to 2025, aggregated for the dashboard. Each row is a group of people who share the same characteristics and place, with counts of how many received each sentence type.
- **Who is counted.** Only people sentenced to prison, because only they can receive supervised release.
- **Exclusions.** The Northern Mariana Islands were left out because there were too few cases.

Sample	
Rows for people sentenced to prison	**487,798**
Rows used in the main analysis (no missing values on the variables in the model)	**459,570 (94.2%)**
Most often missing	**Education (5.5% of rows), race and ethnicity (1.0%), citizenship (0.4%), gender (0.2%)**
Never missing	**Offense category, fiscal year, judicial district**

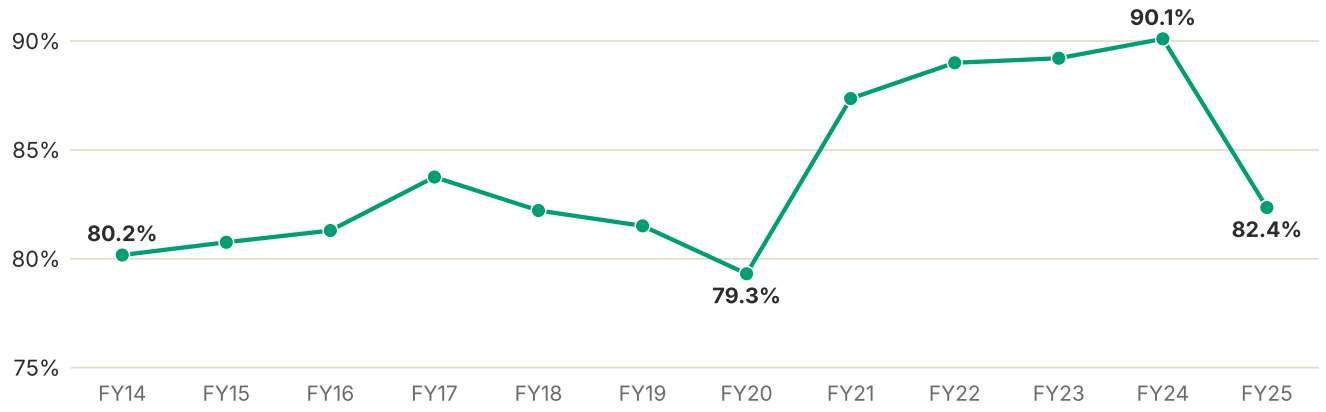


Figure 1. Supervised release rate over time. Share of people sentenced to prison who also received supervised release, national total, fiscal years 2014 to 2025. These are raw rates from the dashboard data. The rate was 80.2% in FY2014, peaked at 90.1% in FY2024 and was 82.4% in FY2025.

2 How to read the results

The outcome was modeled with a binomial mixed-effects (multilevel) logistic regression. In plain terms:

- For each row, the model uses the number of people who received supervised release and the number who did not.
- It estimates how the **odds** of supervised release differ between groups of people, holding the other characteristics in the model constant: age, race and ethnicity, gender, citizenship, education, offense type and fiscal year.
- It gives each state, and each district within a state, its own baseline level, so places can differ even when the people sentenced are alike.
- Results are odds ratios with 95% confidence intervals. An odds ratio of 1 means no difference. Below 1 means lower odds and above 1 means higher odds than the reference group.

Characteristic	Compared with (reference group)
Age	**Each additional year**
Race and ethnicity: Black, Hispanic, Other	**White**
Gender: Female	**Male**
Citizenship: Non-U.S. citizen	**U.S. citizen**
Education: H.S. graduate, Some college, College graduate	**Less than H.S. graduate**
Offense: Firearms, Fraud, Immigration, Sexual, Violent, Other	**Drug trafficking**
Fiscal year	**FY2014**

> **Odds are not percentages.** Odds compare the chance that something happens with the chance that it does not. If 50 in 100 people receive an outcome, the odds are 1, and an odds ratio of 2 doubles them to 2, which is about 67 in 100 people, not 100 in 100. As an illustration, if 98 in 100 U.S. citizens received supervised release (odds of 49), an odds ratio of 0.049 would give non-citizens odds of about 2.4, or about 71 in 100. When an outcome is very common or very rare, a large odds ratio can be a small change in percentage points.

No predictors overlapped strongly (all variance inflation factors were below 1.4). Technical details are in the appendix.

3 What the model found

Who receives supervised release

Figure 2 (below) shows the adjusted odds ratio for every characteristic except age. The sections that follow describe each result.

Citizenship

Non-U.S. citizens had 95% lower odds of supervised release than U.S. citizens who were otherwise similar (odds ratio 0.049, 95% CI 0.047 to 0.051). It is the largest difference in the model. It is consistent with non-citizens being typically subject to deportation rather than supervision in the United States after a prison term.

Offense type

Compared with drug trafficking, sexual offenses had 4.4 times the odds of supervised release, which likely reflects post-prison supervision requirements for these offenses. Other offense types had lower odds: immigration 85% lower (consistent with deportation as the main outcome), other offenses 84%, fraud 61%, violent offenses 54% and firearms 34% lower, although the firearms result is less certain (section 4).

Gender

Women had 13% lower odds of supervised release than men (odds ratio 0.867). For probation, women had higher odds, as the companion report shows.

Race and ethnicity

Compared with White defendants, Hispanic defendants had 17% lower odds (0.828) and defendants in the Other group had 49% higher odds (1.490). Black defendants showed no clear difference from White defendants (odds ratio 1.009, 95% CI 0.947 to 1.076).

Age

Each additional year of age was linked to 0.6% higher odds of supervised release. The effect is small next to the other factors.

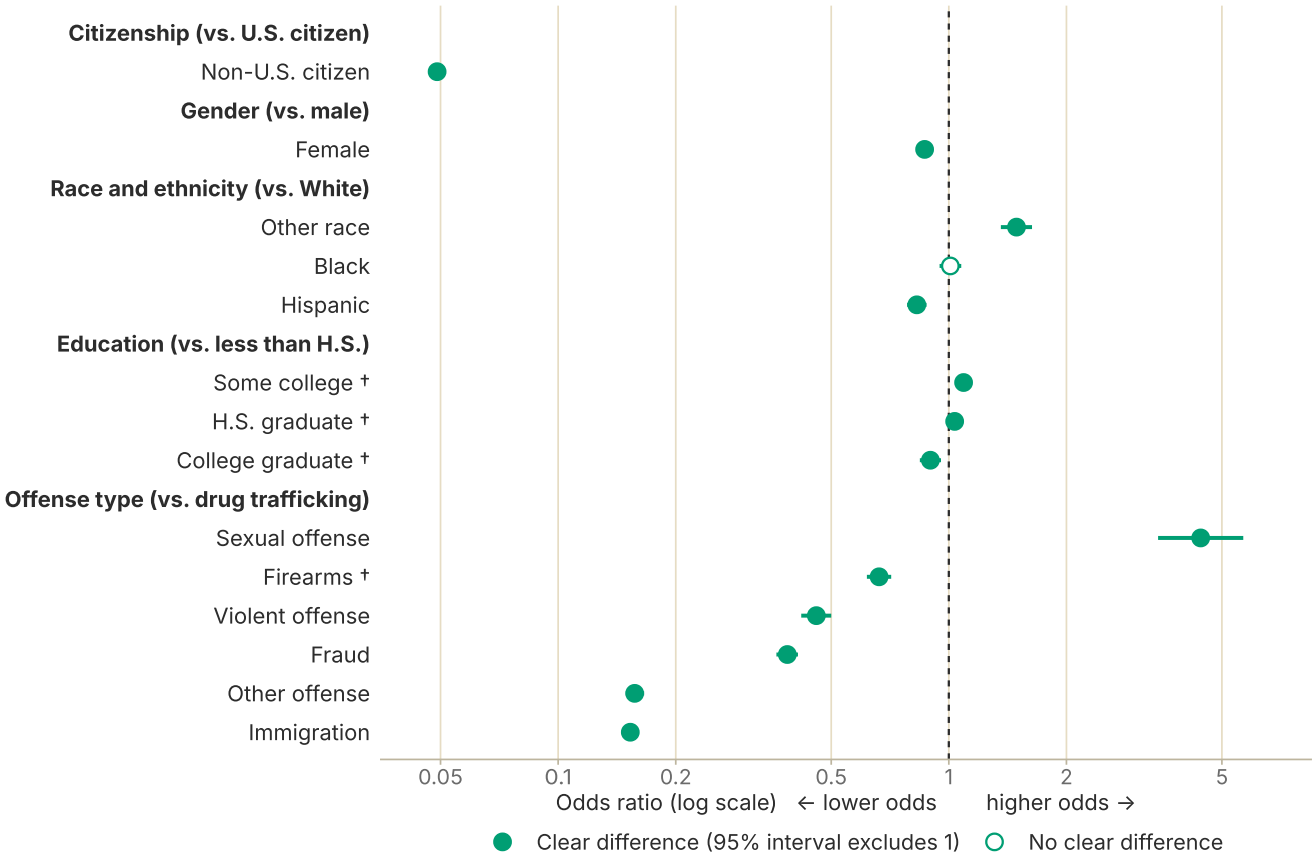


Figure 2. Adjusted odds ratios for supervised release. Odds ratios with 95% confidence intervals from the main model, adjusted for the other factors. Each point compares a group with the reference group named in its row heading. Filled points: the interval excludes 1, so the difference is clear. Hollow point: no clear difference. † Less certain: the result changes when missing data are handled differently (see section 4). The horizontal axis is logarithmic.

Most of the raw Hispanic–White gap reflects citizenship. 72% of Hispanic defendants sentenced to prison were non-citizens, compared with 5% of White defendants. Among U.S. citizens the rates are nearly identical (98.1% for Hispanic and 98.3% for White defendants). Among non-citizens the rate is lower for Hispanic defendants (63.6%) than for White defendants (78.3%). The analysis suggests this gap may reflect different use of deportation and flags it as worth further study.

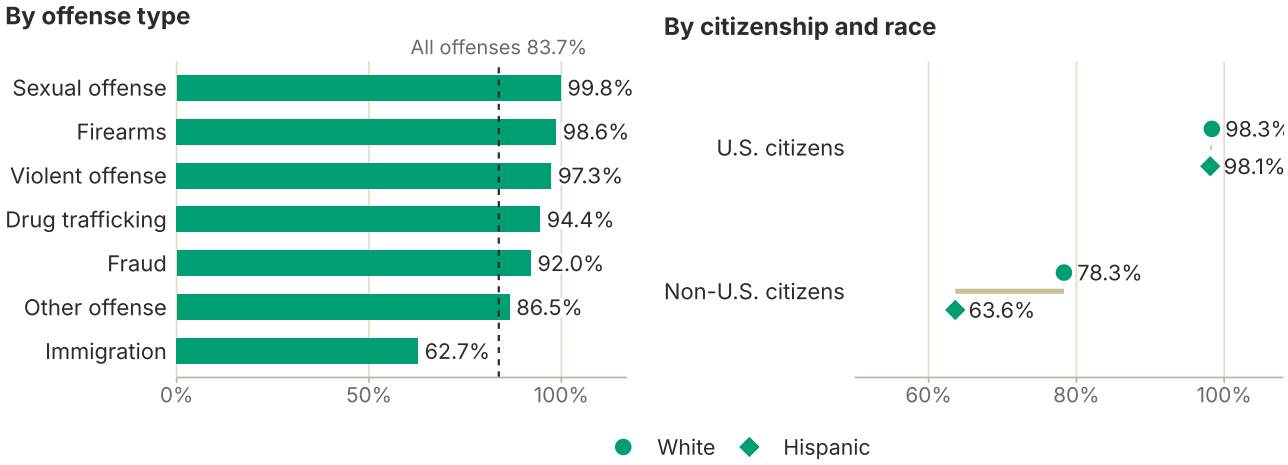


Figure 3. The raw rates behind the odds ratios. Unadjusted supervised release rates among people sentenced to prison, national total, fiscal years 2014 to 2025. They do not hold other factors constant. Right panel: people with a recorded race and citizenship. The dotted line in the left panel is the rate across all offenses.

Education and firearms: less certain

With complete cases, high school graduates and people with some college had 3.5% and 9.1% higher odds than people without a high school diploma, and college graduates had 10% lower odds. That pattern does not hold up. When people with missing education were kept in the analysis, the college graduate result flipped (odds ratio 1.137) or disappeared (1.01, not significant under imputation). People with no education recorded had far lower odds of supervised release (odds ratio 0.088 when missing is treated as its own category), probably because missing education is concentrated among non-citizens and immigration cases.

The firearms result reverses. With complete cases, firearms offenses had 34% lower odds than drug trafficking. With missing as its own category the odds ratio was 1.245, and with multiple imputation 1.44. About 66% of people with missing demographic data had immigration offenses, compared with 31% of people with complete data, so dropping them changes who is being compared. **Treat the education and firearms results as unreliable.**

Change over time

Compared with FY2014, the odds of supervised release were higher in every later year. They rose to FY2017 (1.33), dipped in FY2020 (1.09), which may reflect the pandemic, and then climbed to a peak in FY2024 (1.57). In FY2025 the odds fell to 1.14. Only FY2014 and FY2020 had lower odds than FY2025.

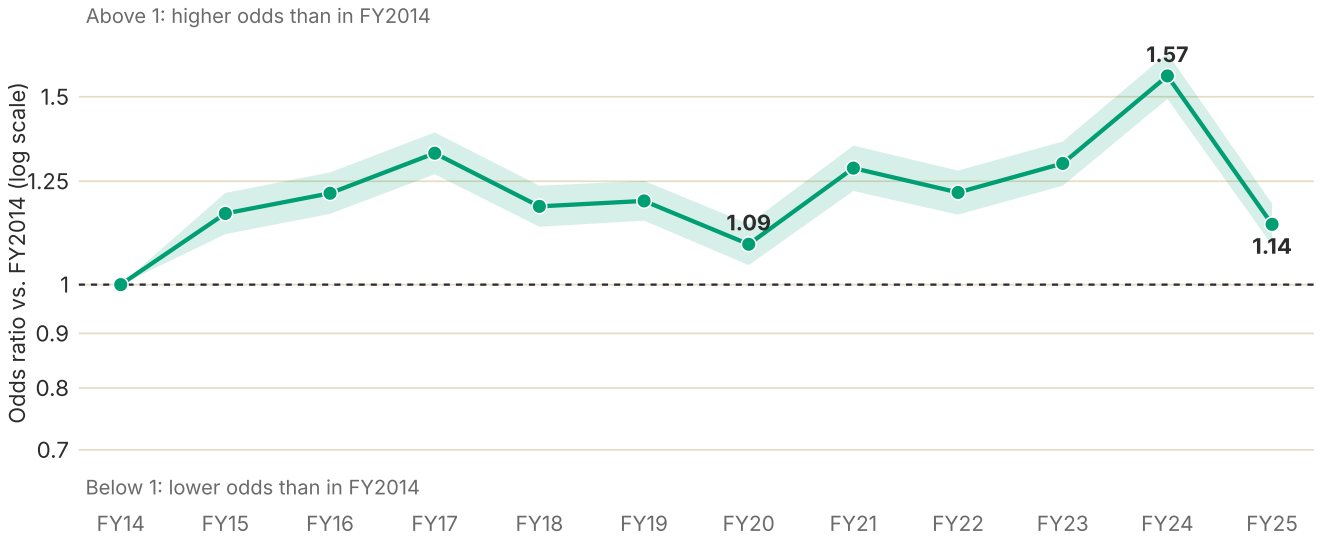


Figure 4. Odds of supervised release by fiscal year. Odds ratio for each fiscal year compared with FY2014, adjusted for the other factors in the model, with 95% confidence bands. The vertical axis is logarithmic.

The FY2025 drop is visible in the raw data too: the supervised release rate fell from 90.1% to 82.4%. It was not matched by an increase in probation, whose rate was 7.2% in FY2025 and 8.1% in FY2024, which suggests more prison-only sentences rather than a shift to other kinds of supervision. The drop held under both missing-data checks. The data cannot show why it happened, and the analysis flags it for further study. A model that replaced the fiscal-year terms with an autoregressive (AR1) structure fit the data much better (AIC lower by 5,267) and left the results for individual characteristics essentially unchanged.

Where someone is sentenced

Supervised release depends heavily on place. In the main model, **31.1%** of the variation left after accounting for people's characteristics is tied to where they were sentenced: 3.5% to the state and 27.6% to the judicial district within the state. Districts differ much more than states: the district-level standard deviation was 1.148, against 0.406 for states.

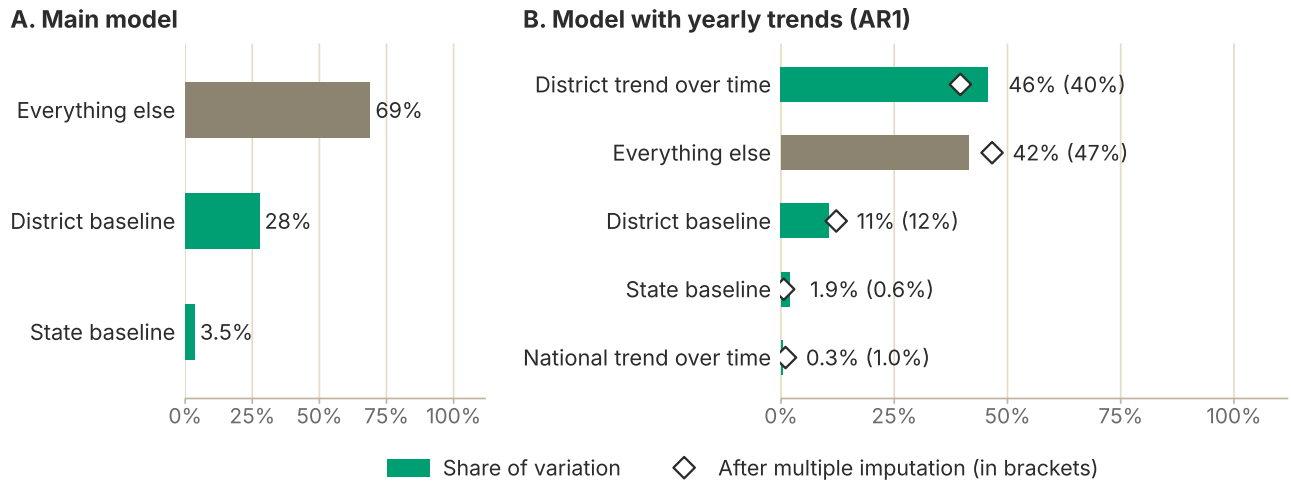


Figure 5. Where the variation in supervised release sits. Share of the total variation tied to each source, on the log-odds scale (intraclass correlation). Panel A is the main model. Panel B lets each district follow its own trend over time (AR1); diamonds show the same model after multiple imputation. The grey bar is everything else: differences between individual cases that place and year do not explain.

When each district can follow its own path over time, the picture sharpens. District-specific trends account for 46% of the variation (40% after imputation) and stable district differences for 11%. Together the district-level share is 56% (52% after imputation), higher than the main model's estimate because that model blends baseline differences with changes over time. National trends account for very little once districts can follow their own paths. Districts differ both in how often they use supervised release and in how that use has changed. The analysis reads this as support for focusing on districts rather than states.

4 How well the findings held up

The main model uses only rows with no missing values (94.2% of rows). Those values are not missing completely at random: education and race are missing far more often for non-citizens and Hispanic defendants (Figure 6), and non-citizens rarely receive supervised release. The pattern fits data that are missing at random given citizenship, the situation that multiple imputation is designed for. Dropping incomplete rows therefore risks selection bias.

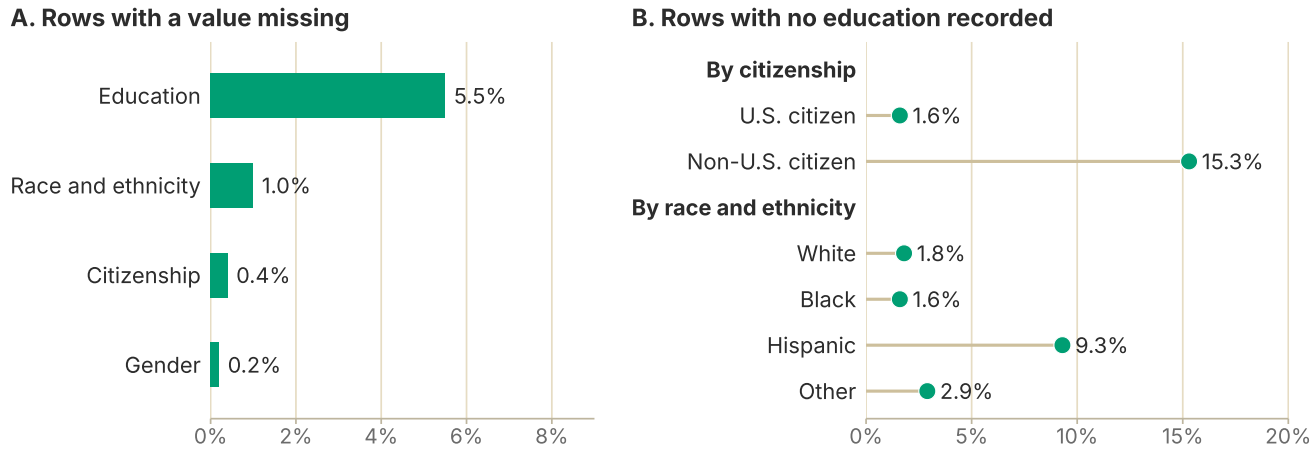


Figure 6. How much was missing, and where. Panel A: share of rows with a missing value for each field. Offense type, fiscal year and district had no missing values. Panel B: share of rows with no education recorded, by group.

The model was therefore run twice more: once keeping people with missing values by treating missing as its own category, and once with multiple imputation (five filled-in datasets using random forests, combined with Rubin's rules).

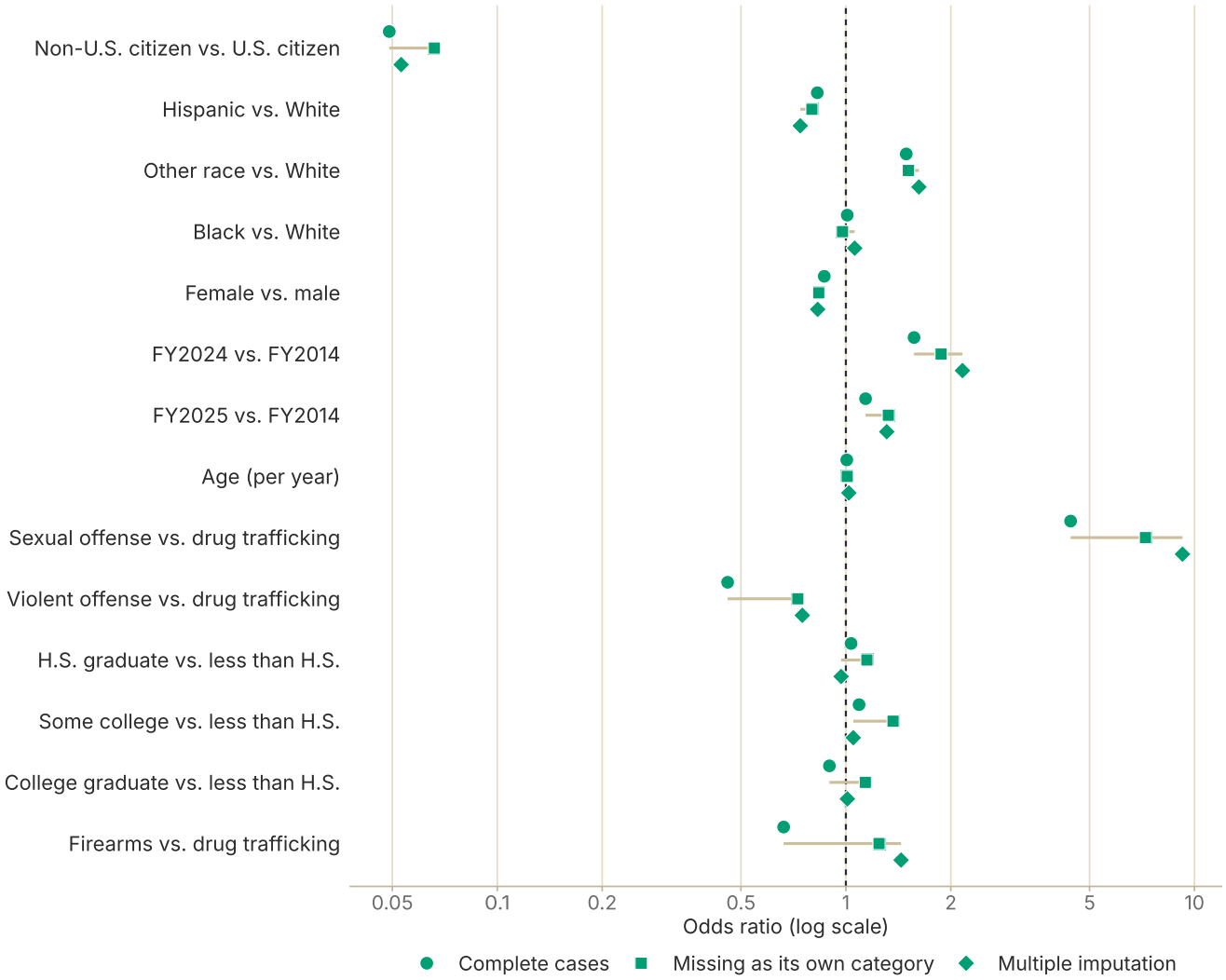


Figure 7. Odds ratios under three approaches to missing data. Odds ratios for the same comparisons from the main model (complete cases), with missing as its own category, and with multiple imputation. Markers that stay close together show a result that holds; markers that spread out or land on the other side of 1 show a result that depends on the method. The horizontal axis is logarithmic.

- **Held:** citizenship, Hispanic and Other race, gender, and the FY2024 and FY2025 results kept their direction and significance. FY2025 stayed well below FY2024 under each approach (odds ratios 1.139, 1.324 and 1.31 for FY2025, against 1.569, 1.876 and 2.16 for FY2024).
- **Size varied:** sexual offenses (4.4 to 9.2) and violent offenses (0.46 to 0.75) kept their direction, but the size moved. Immigration offenses had an odds ratio of 0.153 in the main model and 0.433 under imputation.
- **Did not hold:** the education pattern and the firearms result (section 3).
- **Geography held:** the district-level share of variation was 52% after imputation, against 56% in the main yearly-trend model.

Treating fiscal year as a fixed effect fit better than treating it as a random effect (AIC 181,421.7 against 181,453.8).

5 Limits and open questions

What this analysis cannot tell you

- It describes patterns in recorded outcomes. It does not show why a difference exists, or whether it reflects unequal treatment.

- It leaves out things that influence sentences, such as criminal history, the guideline range, plea agreements and the facts of the case.
- Odds ratios are not differences in percentages. When an outcome is common, as supervised release is, a large odds ratio can be a small change in percentage points.
- The estimates come from one set of data and one set of modeling choices. Confidence intervals are Wald intervals. Profile-likelihood intervals are recommended for final results.

Questions the analysis flags

- Which districts, if any, show the largest racial and ethnic gaps. One approach is to let the race effect vary by district.
- Why Hispanic non-citizens had lower supervised release rates than White non-citizens.
- What explains the drop in supervised release in FY2025.
- Whether the education pattern differs by offense type, for example through models run separately for each type.

How supervised release and probation compare

Comparison	Supervised release	Probation
Non-U.S. citizen vs. U.S. citizen	0.049 (95% lower odds)	0.226 (77% lower odds)
Female vs. male	0.867 (13% lower odds)	2.692 (2.7 times the odds)
Black vs. White	1.009 (no clear difference)	0.701 (30% lower odds)
Hispanic vs. White	0.828 (17% lower odds)	0.775 (23% lower odds)
College graduate vs. less than H.S.	0.897 (10% lower odds)	1.989 (99% higher odds)
Sexual offense vs. drug trafficking	4.413 (4.4 times the odds)	0.163 (84% lower odds)
Fraud vs. drug trafficking	0.386 (61% lower odds)	5.744 (5.7 times the odds)
Immigration vs. drug trafficking	0.153 (85% lower odds)	3.646 (3.6 times the odds)

Adjusted odds ratios from the two reports. The same groups often have opposite results: factors linked to more probation tend to be linked to less supervised release.

Source: U.S. Sentencing Commission individual offender datafiles for fiscal years 2014 to 2025, aggregated for the REFORM Alliance federal sentencing dashboard. Odds ratios are from binomial mixed-effects models fitted with the glmmTMB package in R.

A Appendix: tables and technical details

A1. Odds ratios from the main model

Group	Comparison	Odds ratio	95% CI	p-value	In plain words
Age	Age (per year)	1.006	1.005 to 1.007	<0.001	0.6% higher odds with each year of age
Citizenship	Non-U.S. citizen vs. U.S. citizen	0.049	0.047 to 0.051	<0.001	95% lower odds
Gender	Female vs. male	0.867	0.833 to 0.901	<0.001	13% lower odds
Race/ethnicity	Black vs. White	1.009	0.947 to 1.076	0.777	No clear difference
Race/ethnicity	Hispanic vs. White	0.828	0.782 to 0.877	<0.001	17% lower odds
Race/ethnicity	Other race vs. White	1.490	1.359 to 1.633	<0.001	49% higher odds
Education	H.S. graduate vs. less than H.S.	1.035	1.007 to 1.064	0.014	3.5% higher odds
Education	Some college vs. less than H.S.	1.091	1.046 to 1.138	<0.001	9.1% higher odds
Education	College graduate vs. less than H.S.	0.897	0.843 to 0.955	<0.001	10% lower odds
Offense	Firearms vs. drug trafficking	0.663	0.617 to 0.712	<0.001	34% lower odds
Offense	Fraud vs. drug trafficking	0.386	0.362 to 0.411	<0.001	61% lower odds
Offense	Immigration vs. drug trafficking	0.153	0.148 to 0.158	<0.001	85% lower odds
Offense	Other offense vs. drug trafficking	0.157	0.149 to 0.166	<0.001	84% lower odds
Offense	Sexual offense vs. drug trafficking	4.413	3.433 to 5.671	<0.001	4.4 times the odds
Offense	Violent offense vs. drug trafficking	0.458	0.419 to 0.500	<0.001	54% lower odds

Binomial mixed-effects model, complete cases. Each odds ratio compares a group with its reference group, holding the other characteristics constant.

A2. Fiscal-year odds ratios

Fiscal year	Odds ratio	95% CI	p-value	In plain words
FY2015 vs. FY2014	1.166	1.115 to 1.219	<0.001	17% higher odds than in FY2014
FY2016 vs. FY2014	1.218	1.165 to 1.274	<0.001	22% higher odds than in FY2014
FY2017 vs. FY2014	1.328	1.269 to 1.389	<0.001	33% higher odds than in FY2014
FY2018 vs. FY2014	1.184	1.133 to 1.238	<0.001	18% higher odds than in FY2014
FY2019 vs. FY2014	1.198	1.148 to 1.251	<0.001	20% higher odds than in FY2014
FY2020 vs. FY2014	1.091	1.043 to 1.140	<0.001	9.1% higher odds than in FY2014
FY2021 vs. FY2014	1.286	1.224 to 1.350	<0.001	29% higher odds than in FY2014
FY2022 vs. FY2014	1.220	1.163 to 1.279	<0.001	22% higher odds than in FY2014
FY2023 vs. FY2014	1.299	1.238 to 1.362	<0.001	30% higher odds than in FY2014
FY2024 vs. FY2014	1.569	1.493 to 1.648	<0.001	57% higher odds than in FY2014
FY2025 vs. FY2014	1.139	1.089 to 1.191	<0.001	14% higher odds than in FY2014

A3. Where the variation sits

Source of variation	Main model	Model with yearly trends	Yearly-trend model, multiple imputation
State baseline	3.5%	1.9%	0.6%
District baseline (within state)	27.6%	10.6%	12.2%
State and district baselines combined	31.1%	12.5%	12.8%
District trend over time	–	45.7%	39.6%
National trend over time	–	0.3%	1.0%
District baseline plus district trend	–	56.3%	51.8%
Individual-level variation (remainder)	68.9%	41.5%	46.6%

Shares of total variation on the log-odds scale. The remainder row is one minus the other shares. Yearly-trend model components below.

Component	Std. dev. (complete cases)	Std. dev. (imputed)	AR1 correlation (complete cases)	AR1 correlation (imputed)
State baseline	0.391	0.209	–	–
District baseline	0.917	0.927	–	–
National trend	0.143	0.207	0.515	0.613
District trend	0.921	0.848	0.875	0.861

A4. Results under three approaches to missing data

Comparison	Complete cases	Missing as its own category	Multiple imputation	Result
Non-U.S. citizen vs. U.S. citizen	0.049	0.066	0.053	Holds
Hispanic vs. White	0.828	0.799	0.74	Holds
Other race vs. White	1.490	1.511	1.62	Holds
Black vs. White	1.009	0.978	1.06	Holds: no clear difference
Female vs. male	0.867	0.836	0.83	Holds
FY2024 vs. FY2014	1.569	1.876	2.16	Holds
FY2025 vs. FY2014	1.139	1.324	1.31	Holds
Age (per year)	1.006	1.009	1.02	Same direction, size varies
Sexual offense vs. drug trafficking	4.413	7.236	9.24	Same direction, size varies
Violent offense vs. drug trafficking	0.458	0.728	0.75	Same direction, size varies
H.S. graduate vs. less than H.S.	1.035	1.149	0.969 (not significant)	Changes
Some college vs. less than H.S.	1.091	1.366	1.05 (not significant)	Changes
College graduate vs. less than H.S.	0.897	1.137	1.01 (not significant)	Changes
Firearms vs. drug trafficking	0.663	1.245	1.44	Reverses

Odds ratios. Multiple imputation results for H.S. graduate, Some college and College graduate were not statistically significant.

A5. Model specification

- **Software.** glmmTMB in R. Outcome: `cbind(supervised_release_flag, total_sentenced_fsr - supervised_release_flag)`, binomial family with logit link.
- **Fixed effects.** Age, race and ethnicity, gender, citizenship, education, offense category and fiscal year.
- **Random effects.** Nested random intercepts for judicial district within state, `(1 | state_name/state_district)`.
- **Yearly-trend (AR1) model.** Fiscal-year fixed effects replaced by a national AR1 term, `ar1(0 + year_f | dummy)`, and a district-level AR1 term, `ar1(0 + year_f | state_district)`, keeping the nested intercepts. AIC 176,154.9 against 181,421.7 for fixed-effect years. AR1 correlations: national 0.515, district 0.875 (after imputation 0.613 and 0.861). ICCs for AR1 terms use the variance $SD^2/(1-\rho^2)$.
- **Missing data.** Main model: complete cases. Check 1: missing treated as its own category (all 487,798 rows). Check 2: multiple imputation with miceRanger (random forest, m = 5, 50 trees, 5 iterations, seed 123) for race and ethnicity, gender, citizenship and education; the model was fitted to each imputed dataset and pooled with Rubin's rules.
- **Diagnostics.** All variance inflation factors below 1.4. Missingness was examined with the finalfit package.